Answers and marking guidance

Award one mark per correct, developed point unless stated. Other valid answers should be credited.

2 Complete the truth table for Z.

- A B Z: 0 0 1; 0 1 0; 1 0 0; 1 1 0
- (This is a NOR gate)

3 Quick check

- a) 8
- b) 1
- c) NOT gate
- d) EOR

4 Key terms

- 1 – D
- 2 – F
- 3 – A
- 4 – B
- 5 – E
- 6 – C

5 Write the Boolean equation for Z.

- Z = A.(B + C)
- or Z = A.B + A.C

6 Write the Boolean expression and draw the NAND equivalent with the fewest gates.

- Z = A.B + C
- Each gate replaced by its NAND equivalent
- Double inversions cancelled to give the fewest gates